

Flow simulations

We can see 2D Streamlines and 3D flow trajectories for steady state: we can select the results to show based on velocity, pressure, temperature, and many others.

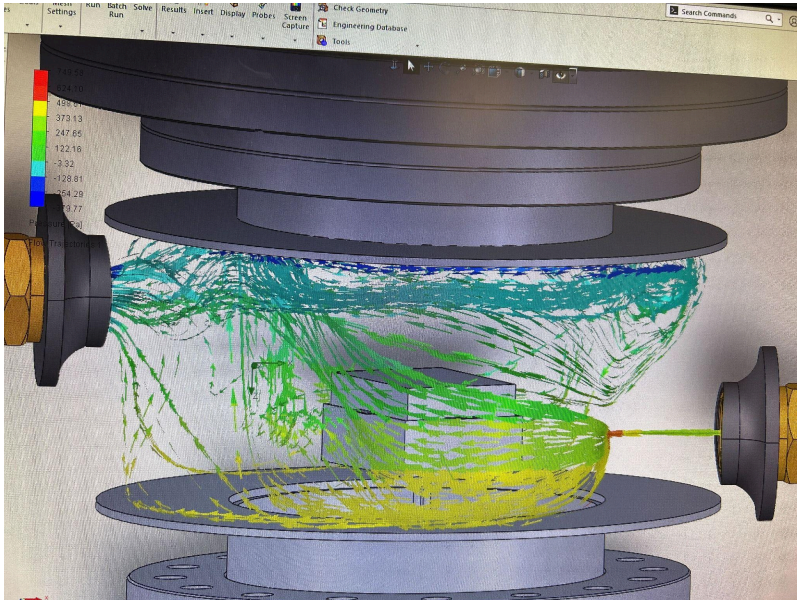
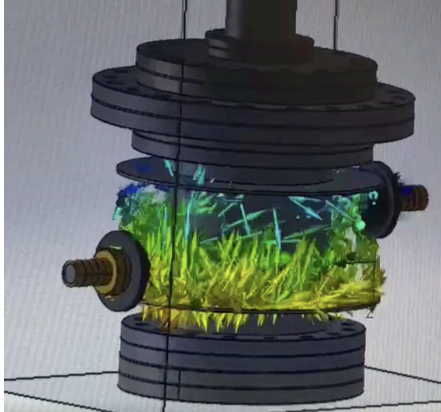
Issues:

Too many parameters to be set

Conditions have to be in contact with the water chamber: had to set the inner chamber outer wall to 1200 °C

My mistakes:

Initial results looked unpleasant: water didn't fill up the entire region:



I found out that I initially selected the inlet and outlet as the entire hose fittings, but I only had to select the plane that is perpendicular to the water flow.

1/ Dummy Flow Simulations: 1 pipe (to compare solidworks and theory)

I did not use fluid subdomains as I let solidworks calculate that for me to make sure it matches theory

Pressure Drop Online-Calculator

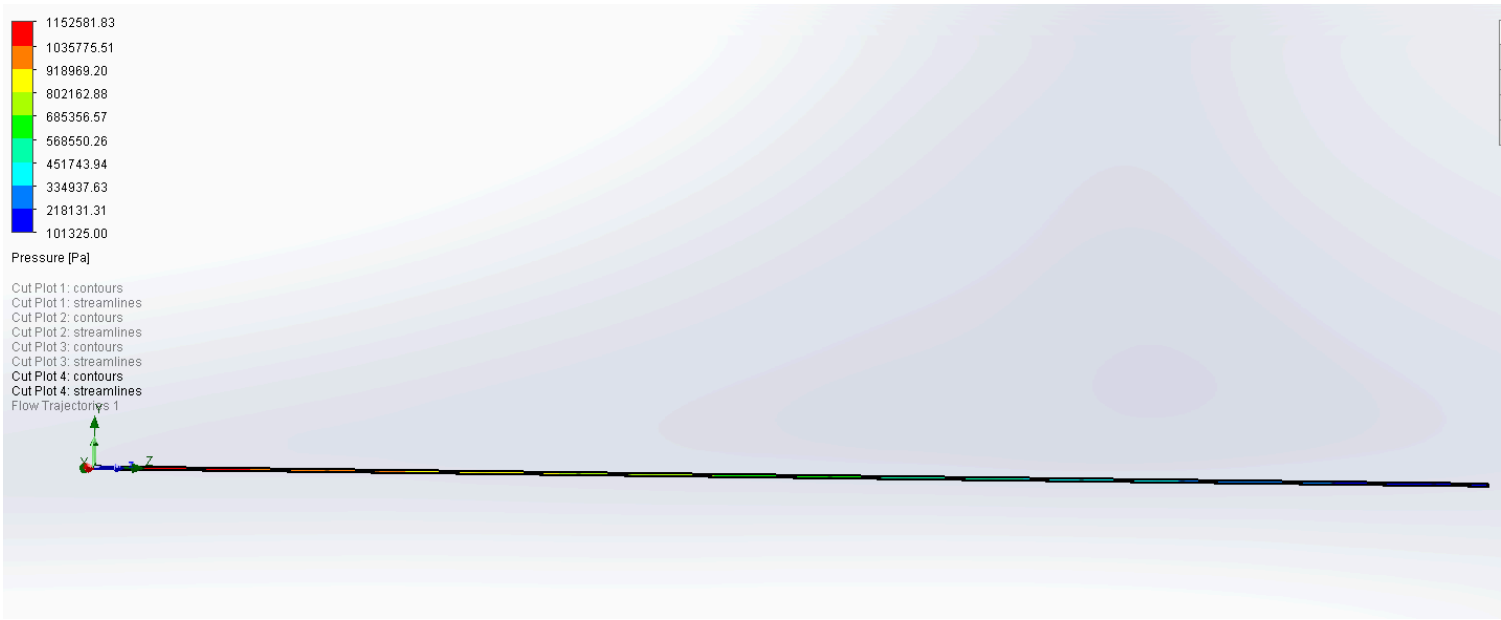
Calculation output

Flow medium:	Water 20 °C / liquid
Volume flow:	20 m3/h
Weight density:	998.206 kg/m3
Dynamic Viscosity:	1001.61 10-6 kg/ms
Element of pipe:	circular
Dimensions of element:	Diameter of pipe D: 20 mm
	Length of pipe L: 10 m
Velocity of flow:	17.68 m/s
Reynolds number:	352476
Velocity of flow 2:	-
Reynolds number 2:	-
Flow:	turbulent
Absolute roughness:	0 mm
Pipe friction number:	0.01
Resistance coefficient:	7.02
Resist.coeff.branching pipe:	-
Press.drop branch.pipe:	-
Pressure drop:	10950.24 mbar
	10.95 bar

Note: The pressure drop was calculated by the online calculator of www.pressure-drop.com. We can not warrant the correctness of this software. The software is produced carefully. But no computer software is without bugs. Therefore the calculations are your own risk.

Do you know our software SF Pressure Drop for Excel or for Windows? www.pressure-drop.com

<https://www.pressure-drop.com/Online-Calculator/dp.php>



Results:

Parameter	Value
Flow rate	0.00556 m³/s (20 m³/h)
Pipe length	10 m
Diameter	20 mm (0.02 m)

Outlet pressure	1 atm = 101325 Pa
Inlet pressure (from simulation)	1,152,581 Pa
Expected ΔP (from online calculator)	10.95 bar = 1,095,024 Pa
Resulting ΔP (from SolidWorks)	1,152,581 – 101,325 = 1,051,256 Pa

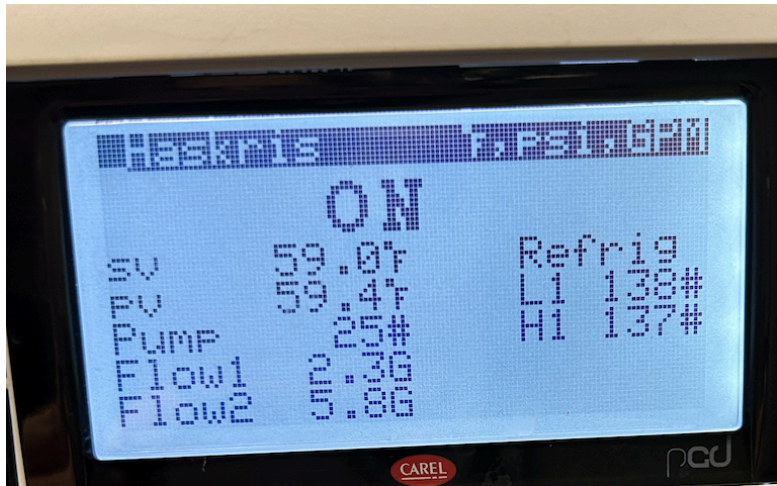
% error = 4% (very reasonable)

Compared to previous work that I have done myself, I see that inconsistency was due to setting fluid subdomains myself (because I would have to include the fluid pressure). Therefore, as long as we let solidworks choose the fluid volume, simulations will almost match theory.

2/ Simple Flow Simulations: only the water pipe part in our design:

Please note that this will probably be the most accurate simulation since solidworks determines the water pressure.

For the complex design, I have to manually add the water pressure.



This is our chiller information:

Parameter	Value
Pump pressure	25 psi (25#)
Flow 1	2.3 GPM = 0.145 L/s
Flow 2	5.8 GPM = 0.367 L/s
Refrigerant L1	138 psi
Refrigerant H1	137 psi
Hose diameter	5/8 inch
Inlet Pressure	= 25 PSI = 172369 Pa

We will calculate flow rate

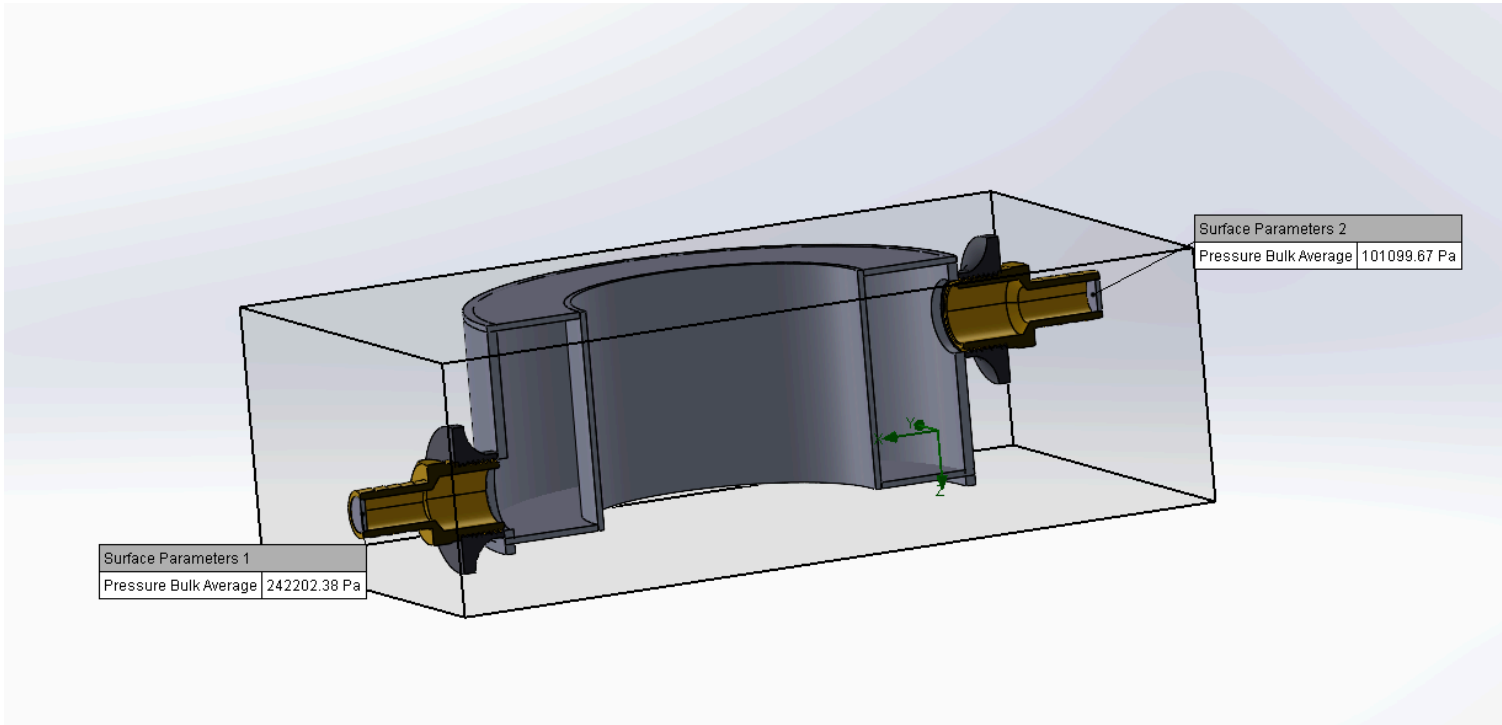
$Q = A \cdot v$
 $A = \pi(d/2)^2$

$v_1 = 0.73\text{m/s}$

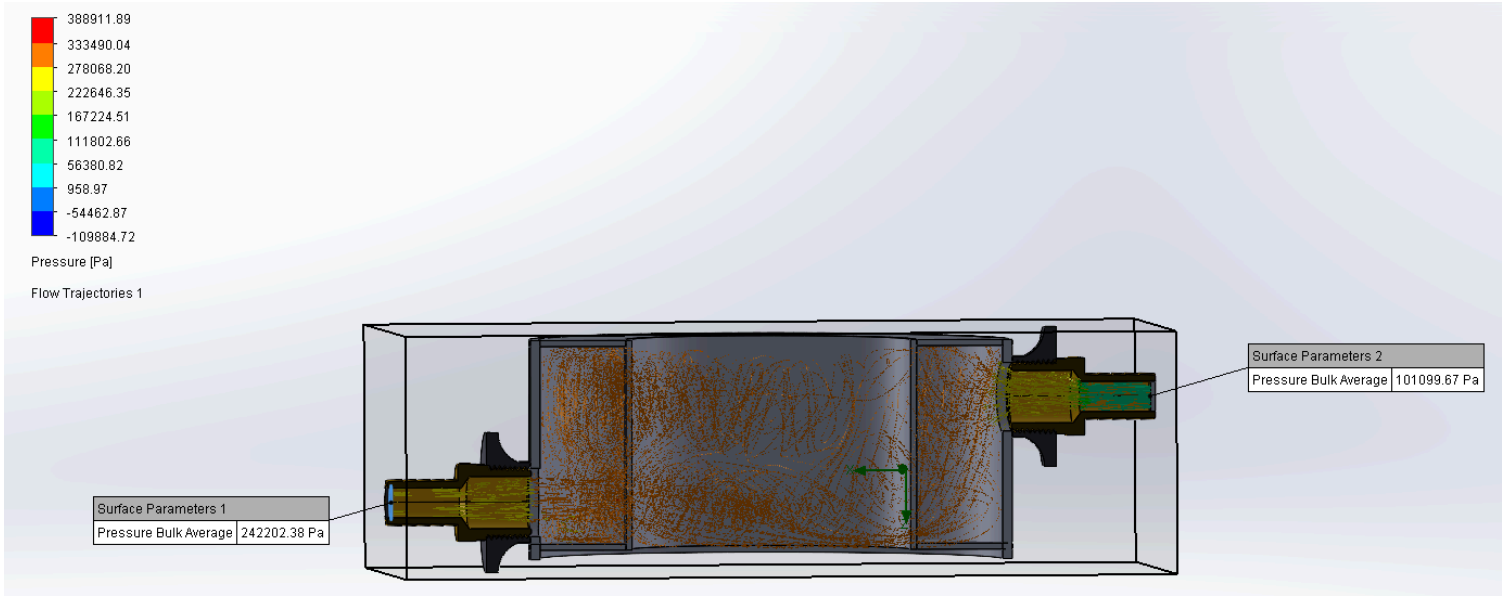
$v_2 = 1.85\text{m/s}$

Pressure Drop = $P(\text{inlet}) - P(\text{outlet}) = 242202.38 - 101099.67 = 141102.71 \text{ Pa}$

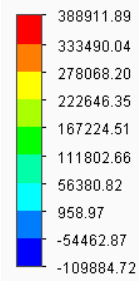
Pressure Drop:



Pressure Flow Trajectories

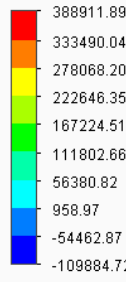
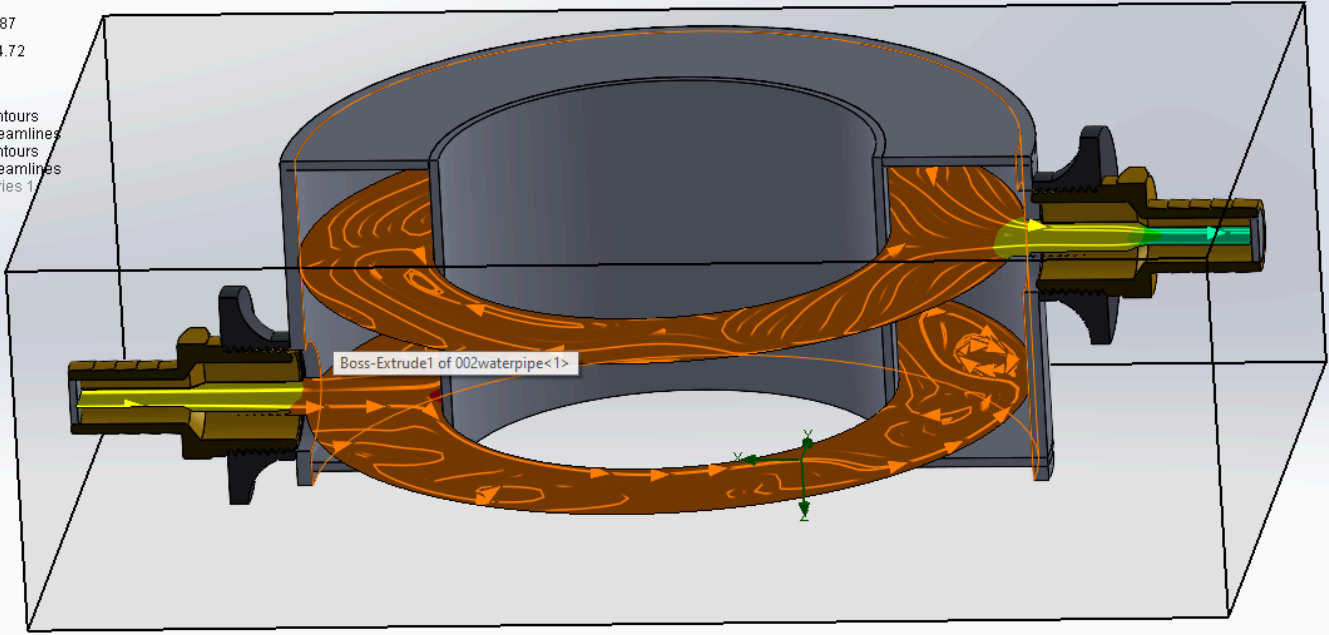


Pressure streamlines:



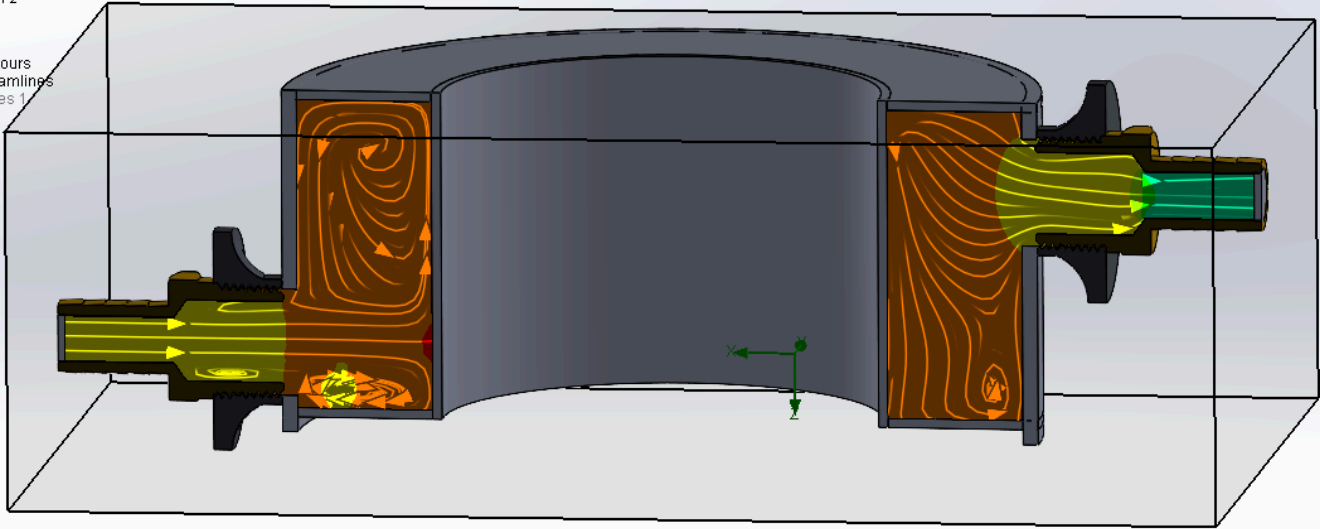
Iteration = 343

Pressure [Pa]
Cut Plot 1: contours
Cut Plot 1: streamlines
Cut Plot 2: contours
Cut Plot 2: streamlines
Flow Trajectories 1

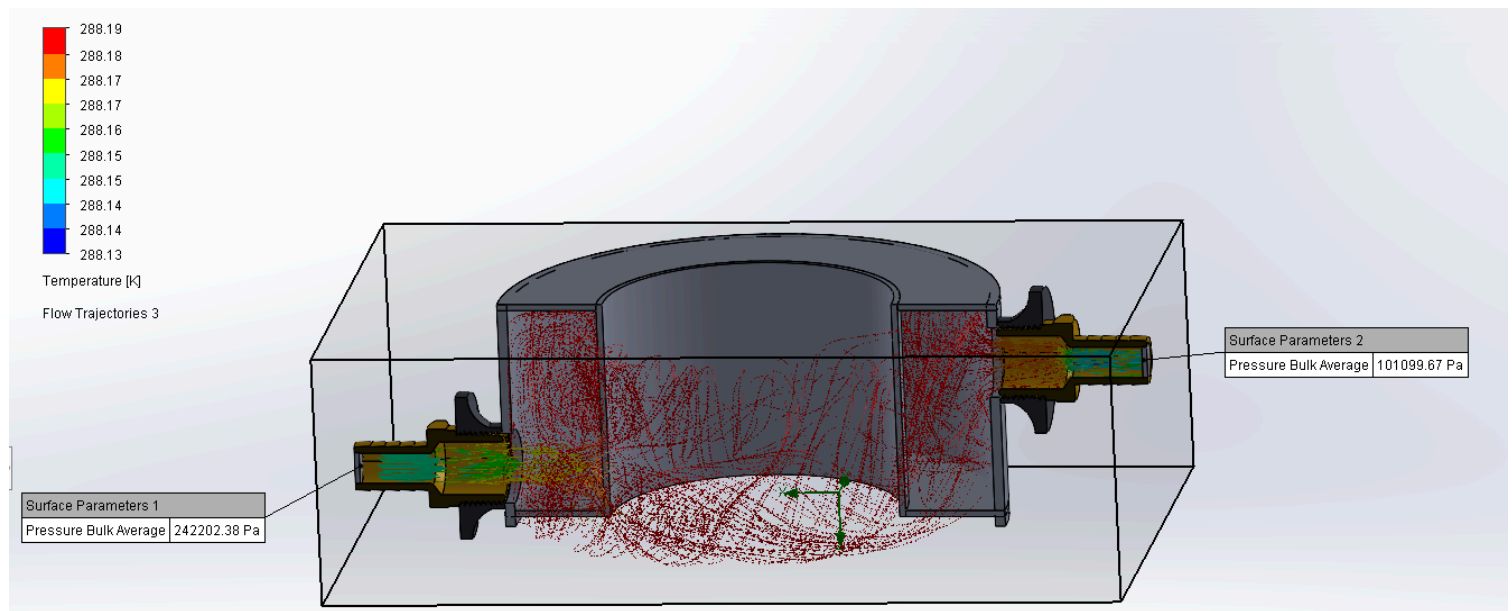


Iteration = 343

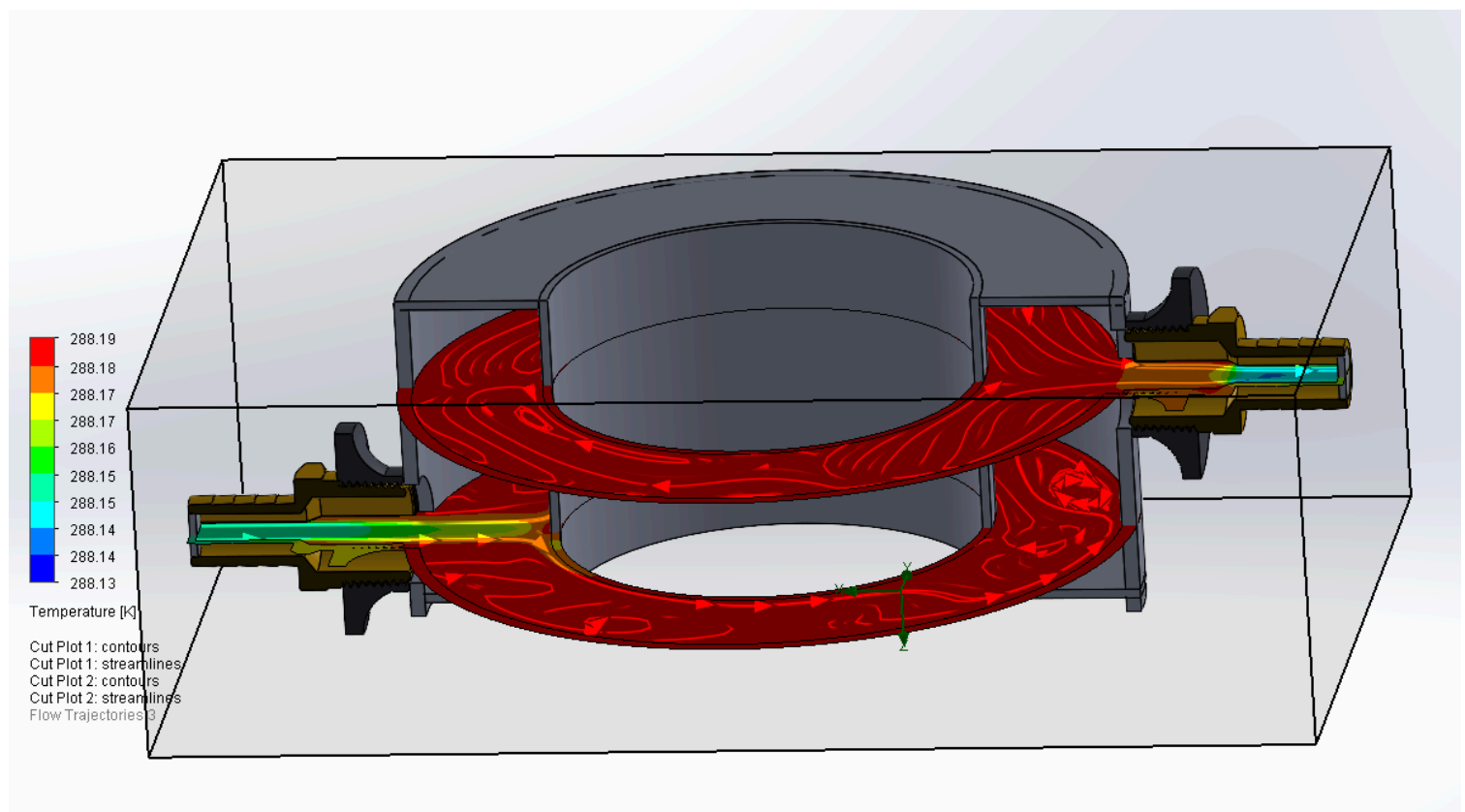
Pressure [Pa]
Cut Plot 3: contours
Cut Plot 3: streamlines
Flow Trajectories 1

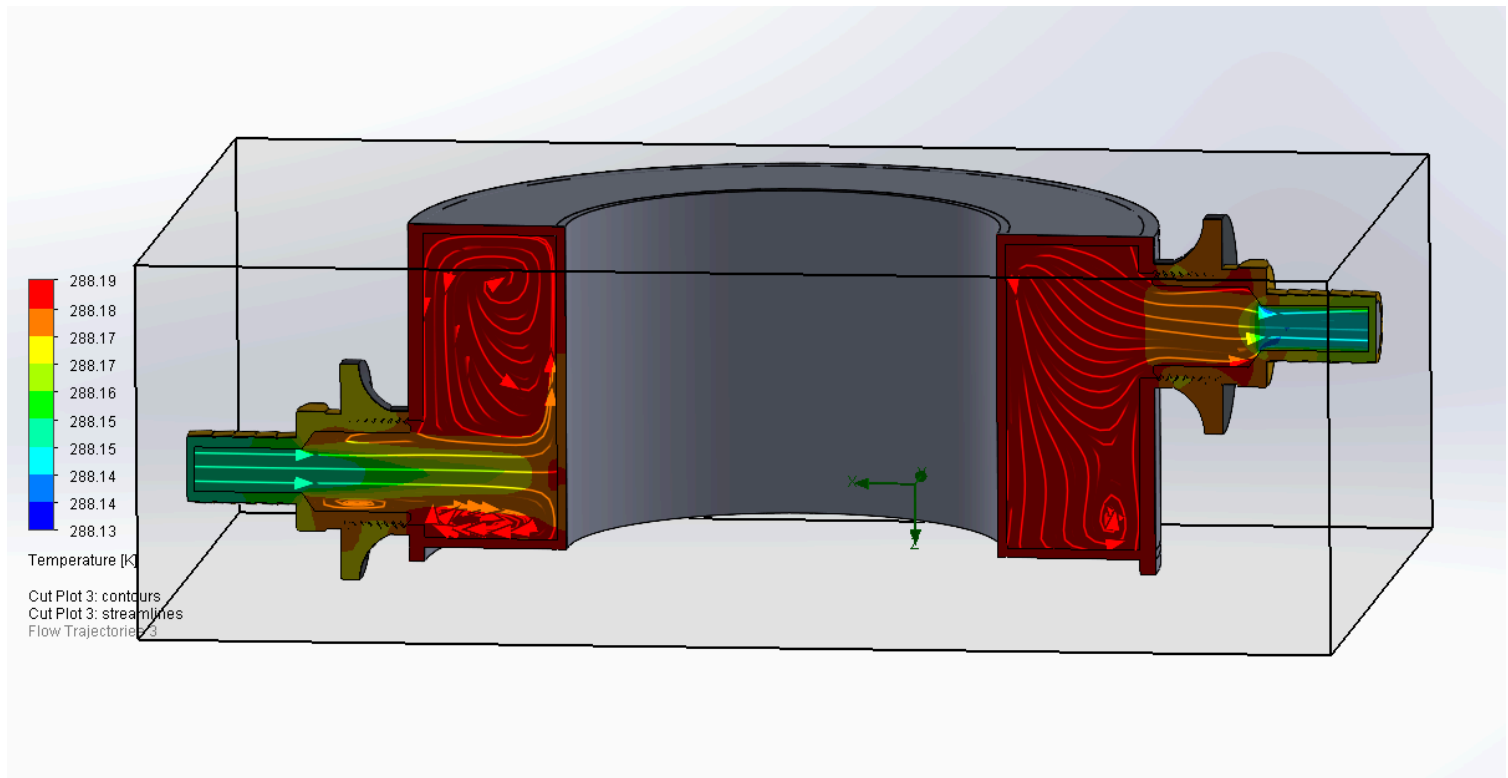


Temperature Flow trajectories:

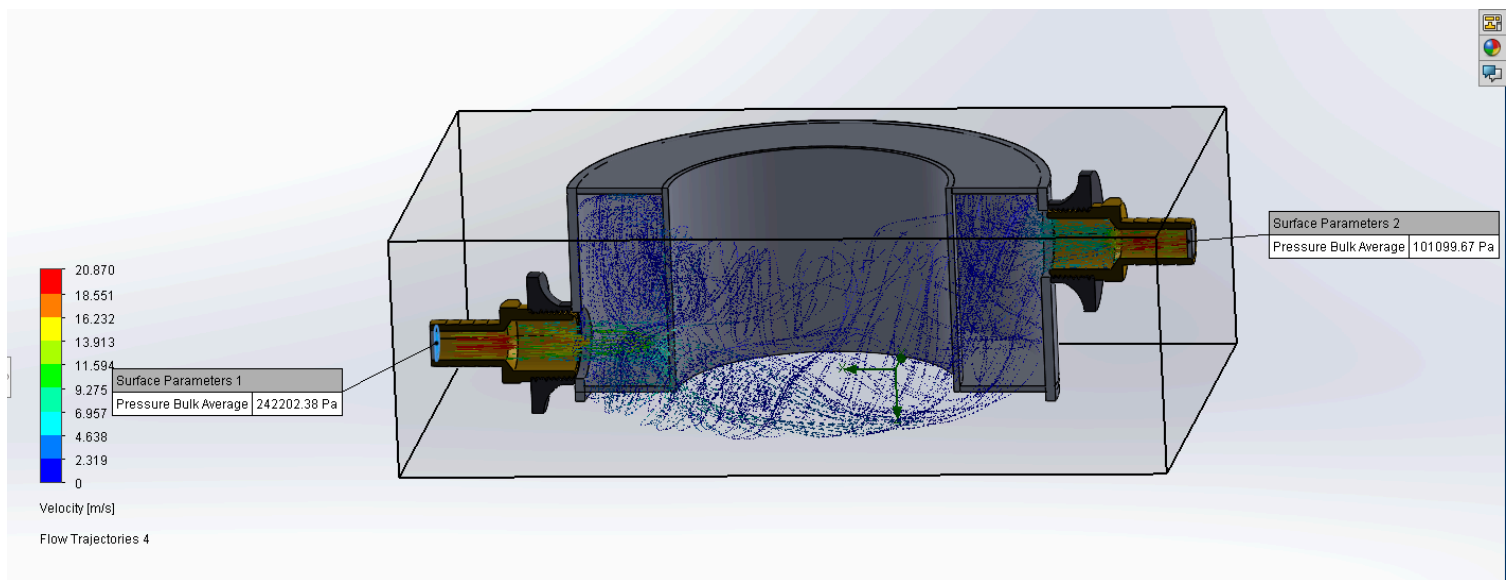


Temperature streamlines:

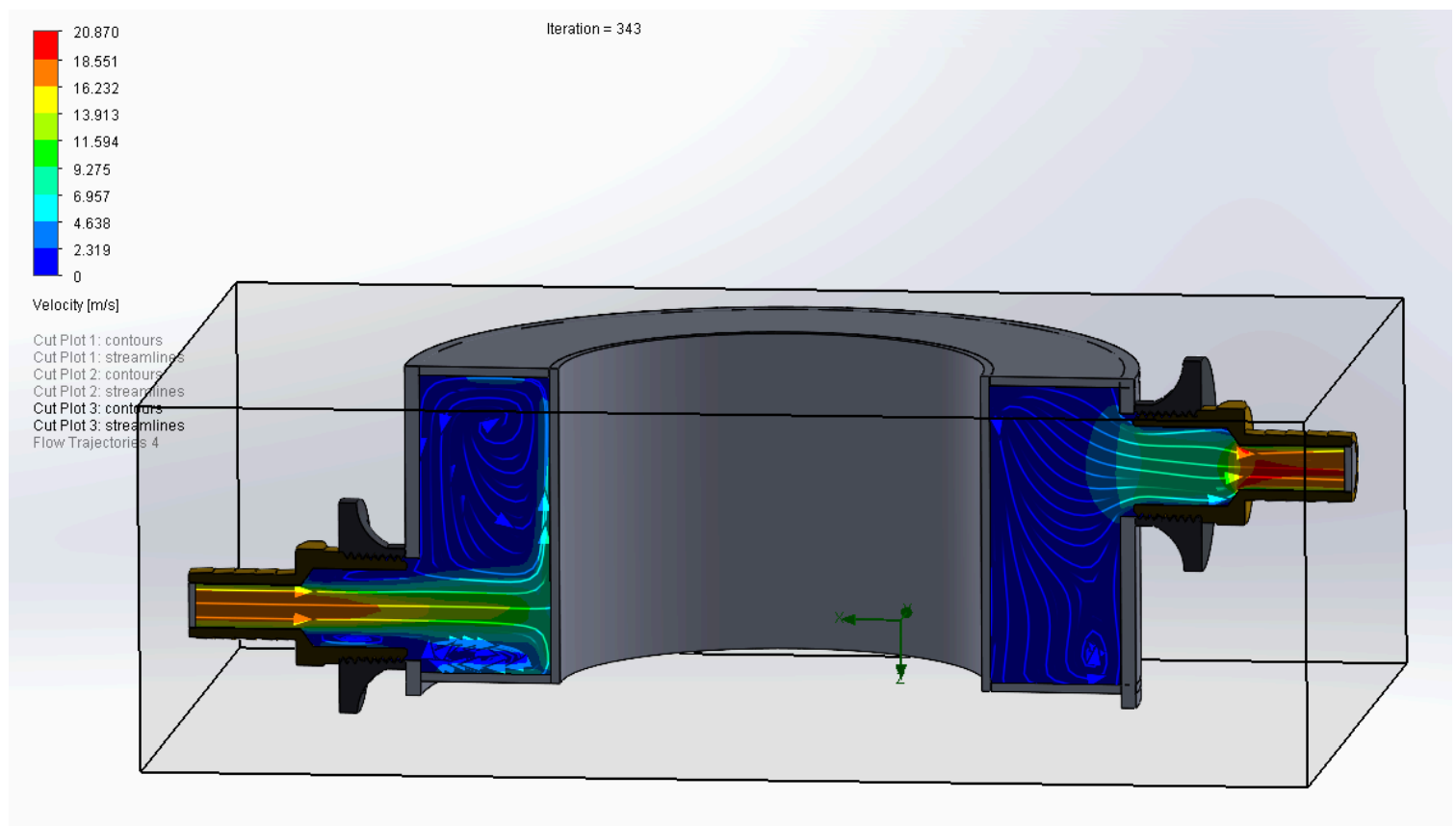
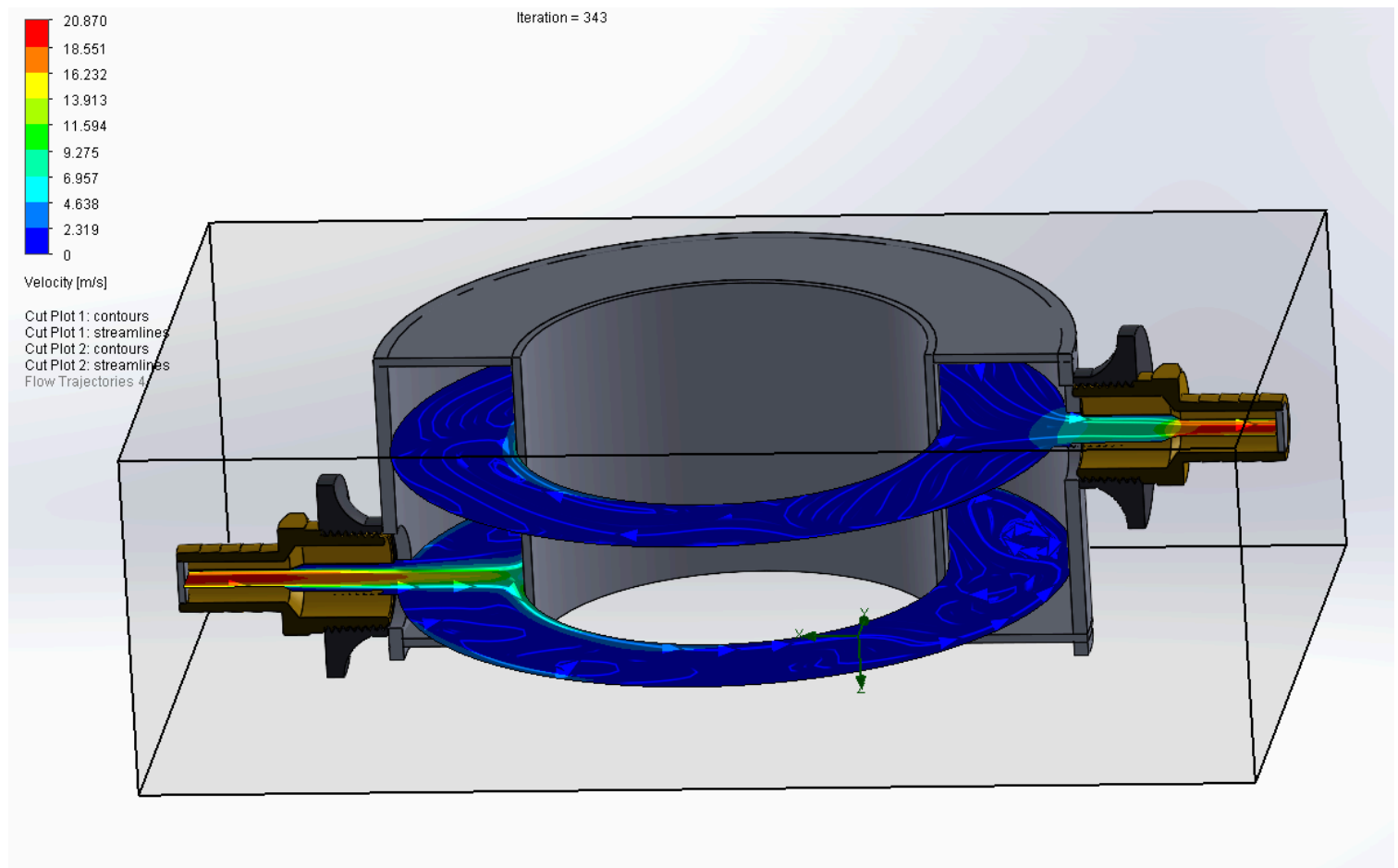




Velocity flow trajectories

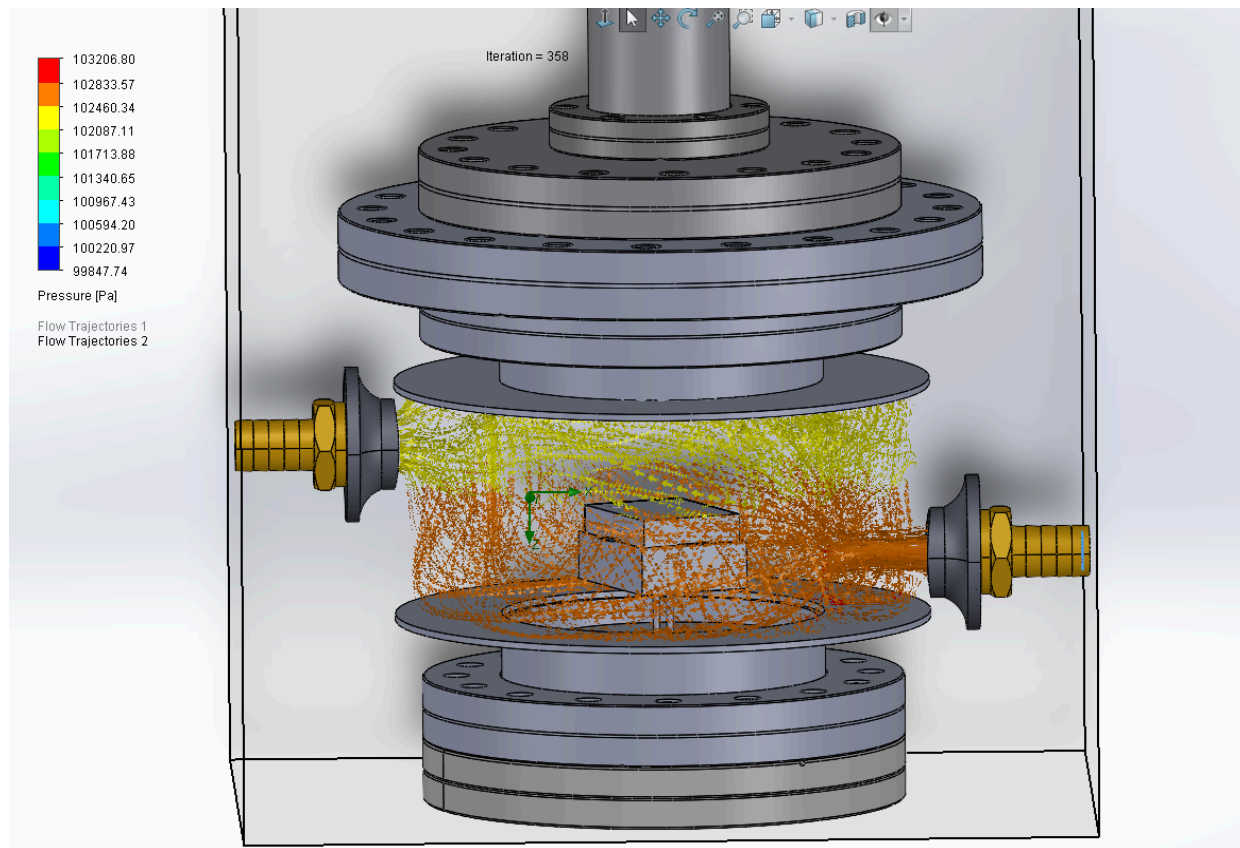


Velocity streamlines:

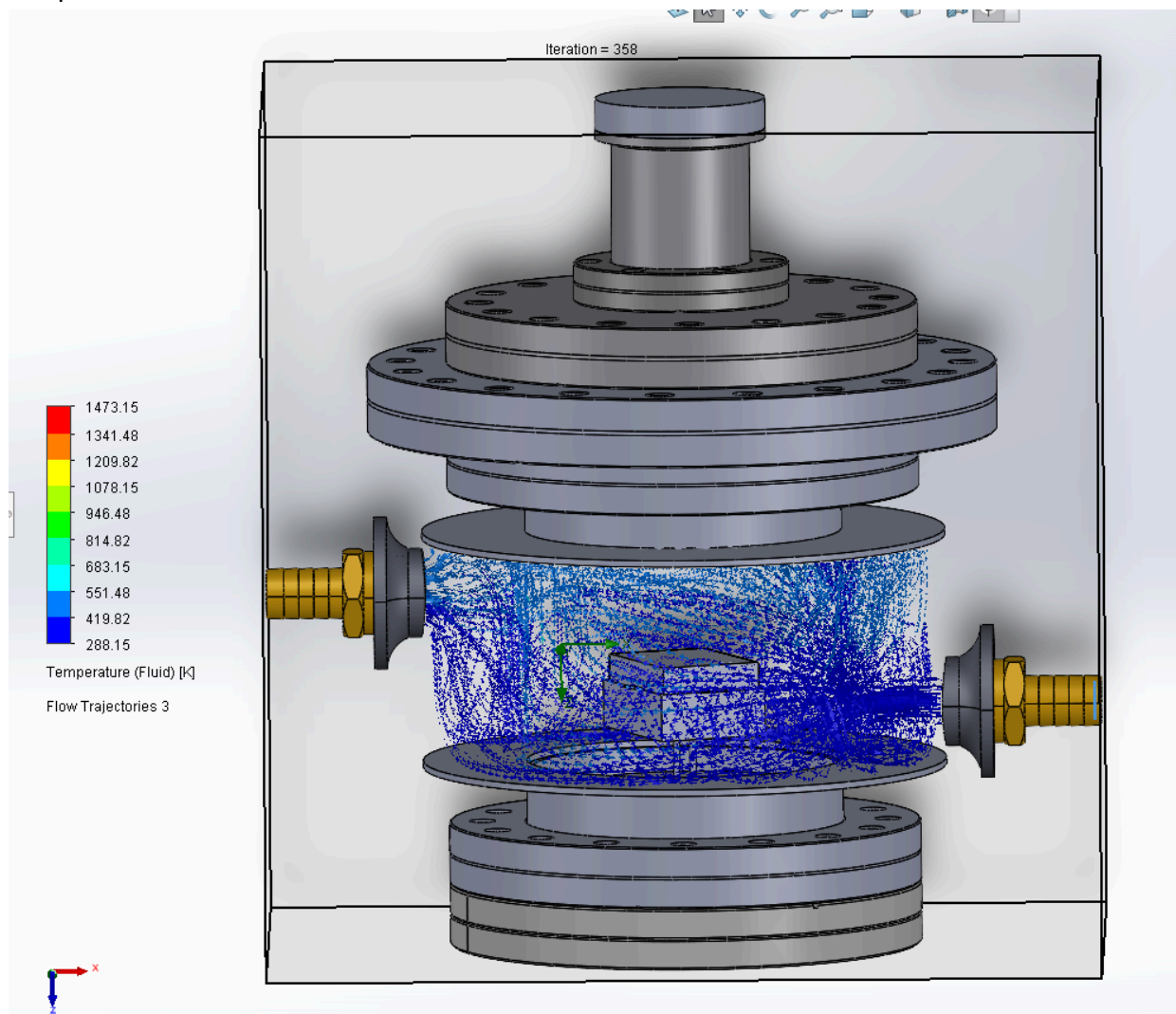


3/ Complex Flow Simulations:

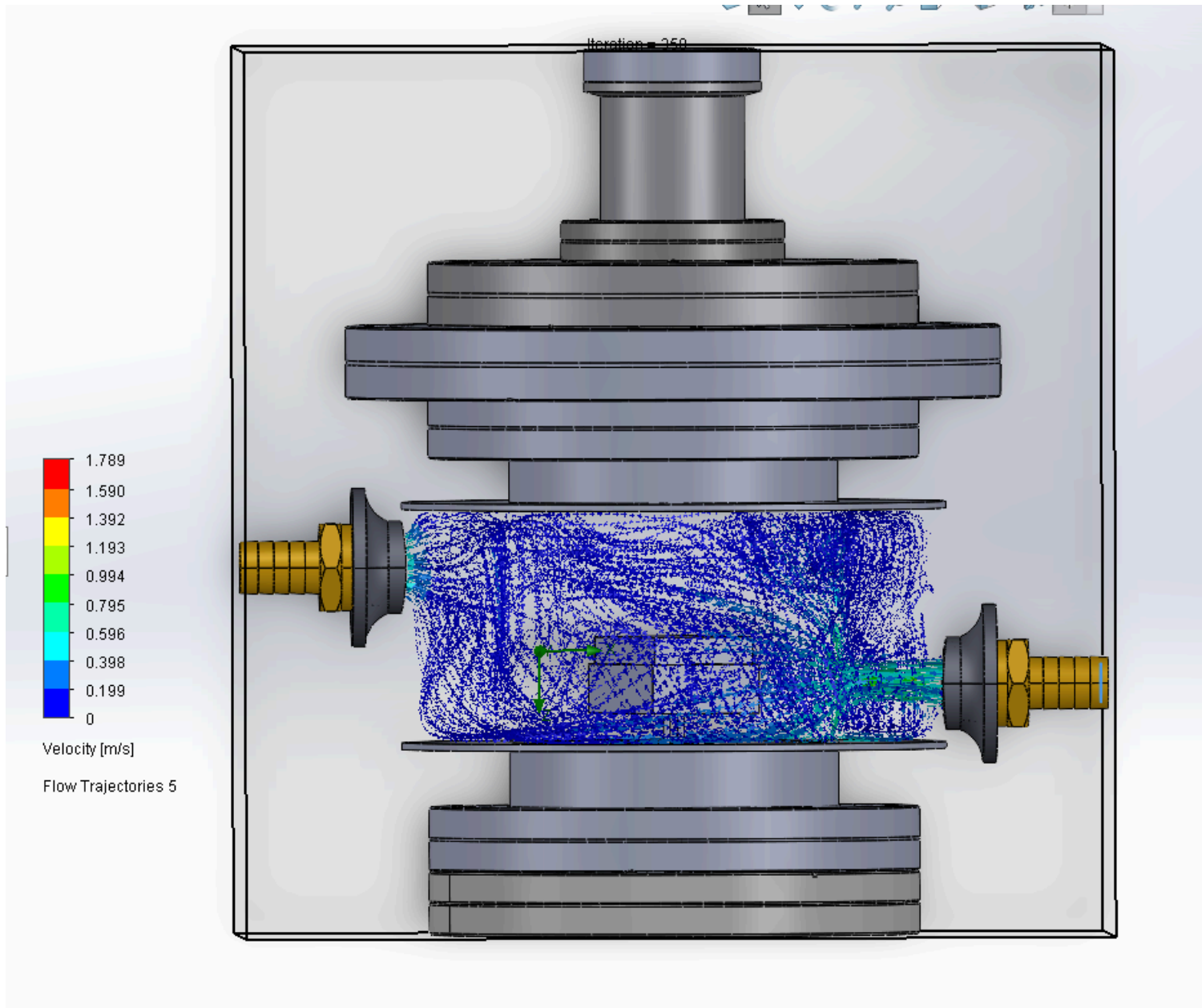
Pressure Results:



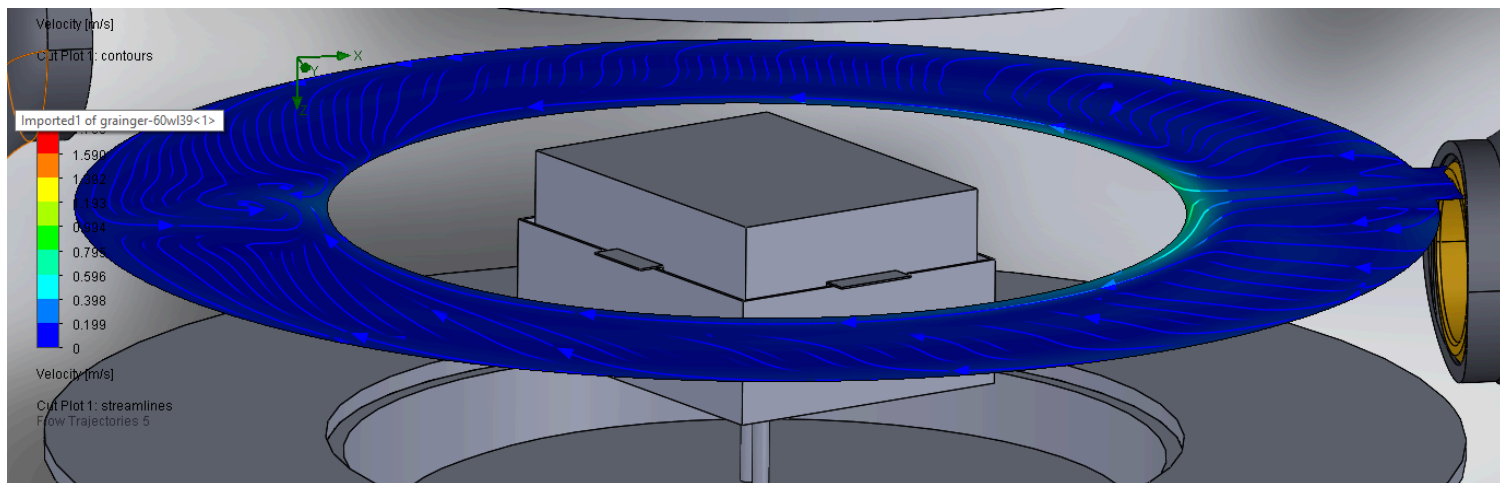
Temperature results:

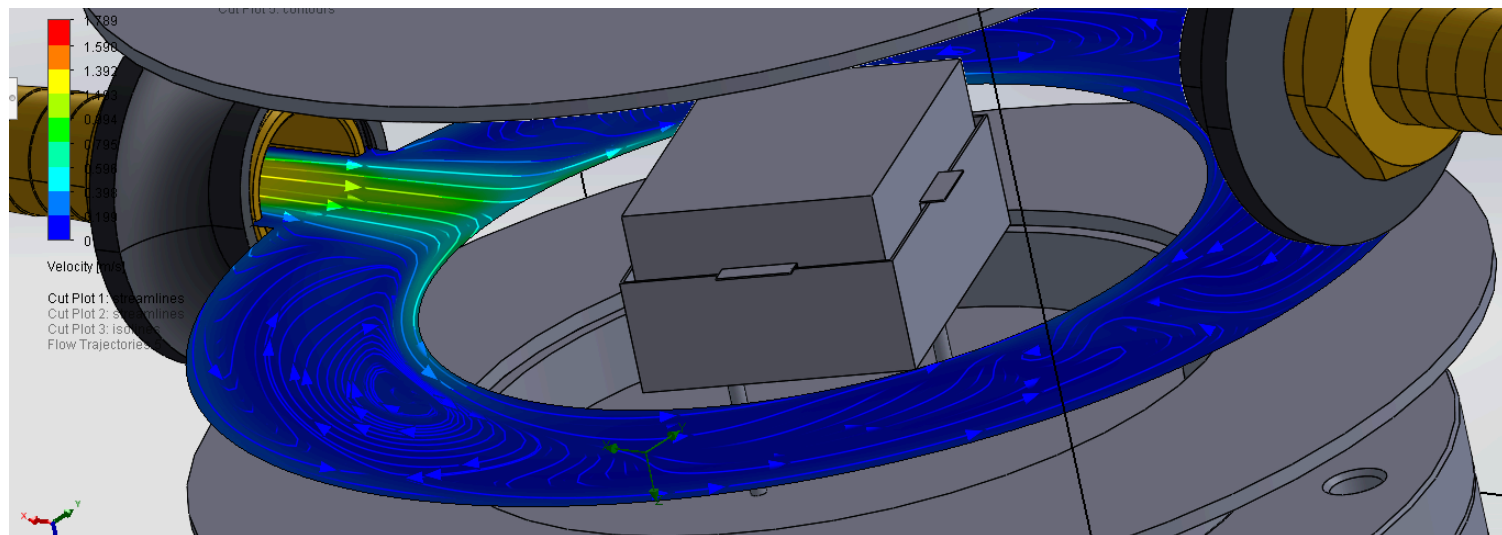


Velocity results:



Horizontal Streamlines:





Vertical streamlines:

